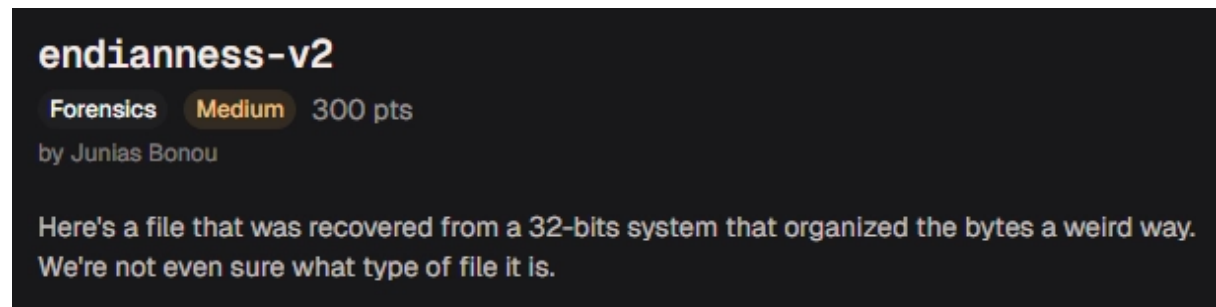


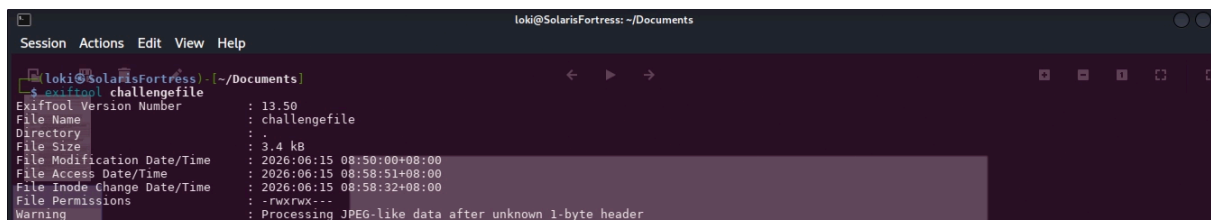
endianness-v2 || picoCTF Writeup

written by Alexander Sapo

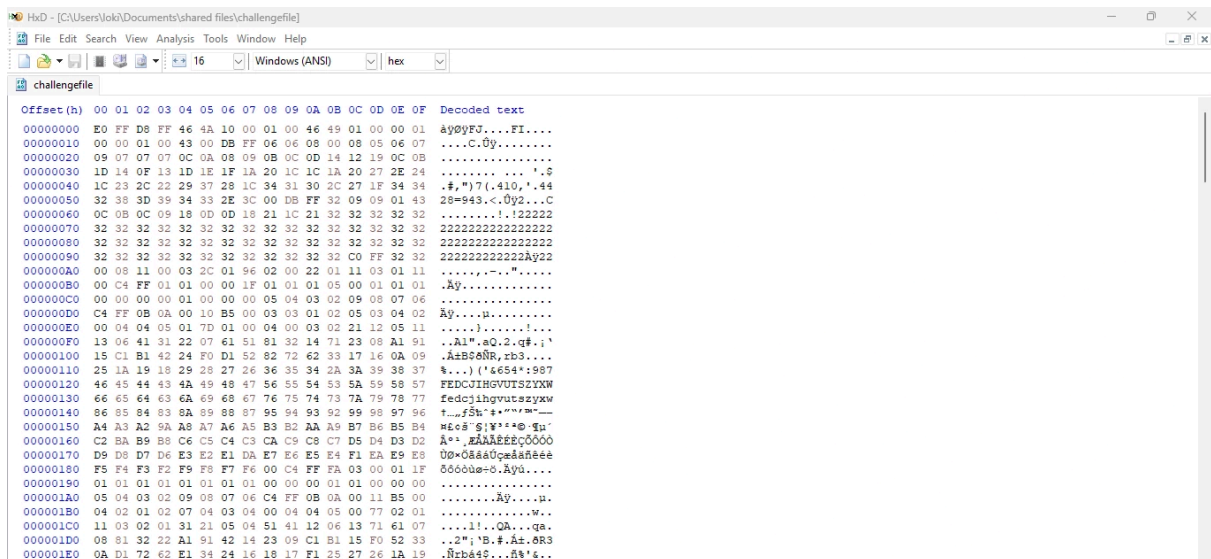


process:

As mentioned earlier, the recovered file originated from a 32-bit system where the data was stored as 32-bit little-endian words instead of the original file byte order. We first ran `exiftool`, which identified the file as a JPEG image.

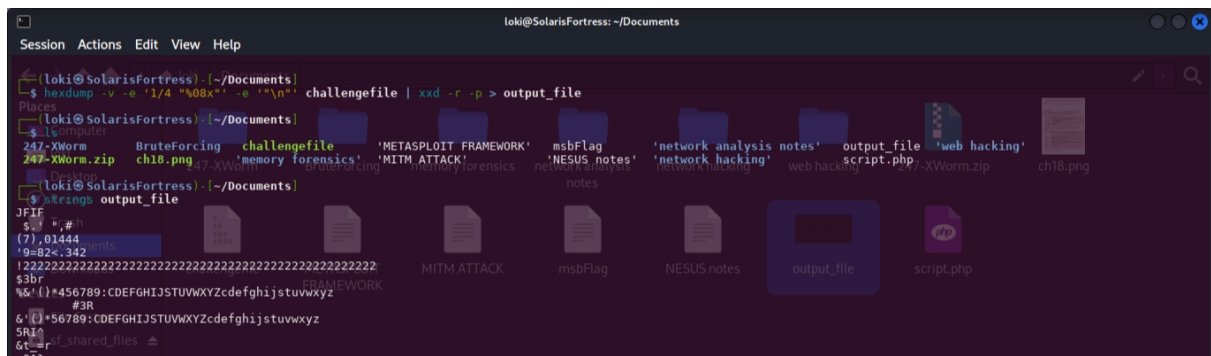


To verify this, we examined the file using a hex editor and noticed that the JPEG magic bytes appeared as `E0 FF D8 FF 46` rather than the expected `FF D8 FF E0 00 10 4A 46`. We also checked the file footer, which matched the expected JPEG end marker, further confirming that the recovered file was indeed a JPEG image.



After researching the unusual byte ordering, we found that each 4-byte word had been stored in little-endian format. As a result, the file could not be interpreted correctly until the byte order of every 4-byte block was restored. To reconstruct the original JPEG, we used the following command:

```
hexdump -v -e '1/4 "%08x"' -e '"\n"' challengefile | xxd -r -p > output_file
```



This command interprets each 4-byte block as a 32-bit little-endian value and outputs it in the correct byte order. The resulting hexadecimal stream is then converted back into a binary file using `xxd -r -p`, producing a valid JPEG image. As an alternative, the same result can be achieved in CyberChef by using the **Swap Endianness** operation with a **4-byte word length**, which restores the correct byte order and allows the image to be rendered successfully.

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Operations 481

- render
- Render Image
- Render Markdown
- Fang URL
- HTML To Text
- Hex Density chart
- Template
- To Table
- Favourites
- Data format
- Encryption / Encoding
- Public Key
- Arithmetic / Logic

Recipe

- From Hex
 - Delimiter: Auto
- Swap endianness
 - Data format: Raw
 - Word length (bytes): 4
 - ☐ Pad incomplete words
- Render Image
 - Input format: Raw

STEP

BAKE!

Auto Bake

Input

```

E0 FF D8 FF 46 4A 10 00 01 00 46 49 01 00 00 01 00 00 01 00 43 00 D8 FF 06 06 08 00
08 05 06 07 09 07 07 07 0C 0A 08 09 08 0C 0D 14 12 19 0C 08 10 14 0F 13 1D 1E 1F 1A
20 1C 1C 1A 20 27 2E 24 1C 23 2C 22 29 37 28 1C 34 31 30 2C 27 1F 34 34 32 38 3D 39
34 33 2E 3C 00 D8 FF 32 09 09 01 43 0C 08 0C 09 18 00 0D 18 21 1C 21 32 32 32 32 32
32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32
32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32
02 00 22 01 11 03 01 11 00 C4 FF 01 01 00 00 1F 01 01 01 05 00 01 01 01 00 00 00
01 00 00 00 05 04 03 02 09 08 07 06 C4 FF 08 0A 00 10 B5 00 03 03 01 02 05 03 04
00 04 04 05 01 7D 01 00 04 00 03 02 21 12 05 11 13 06 41 31 22 07 61 51 81 32 14
23 08 A1 91 15 C1 B1 42 24 F0 D1 52 82 72 62 33 17 16 0A 09 25 1A 19 18 29 28 27
36 35 34 2A 3A 39 38 37 46 45 44 43 4A 49 48 47 56 55 54 53 5A 59 58 57 66 65 64 63

```

10262
1
Raw Bytes

Output

```

picoCTF(cert!f1Ed_iNd!4n_s0rrY_3nDian_76e05f49)

```

flag:

output_file - Image Viewer [2/2]

File Edit View Go Help

← →




picoCTF(cert!f1Ed_iNd!4n_s0rrY_3nDian_76e05f49)